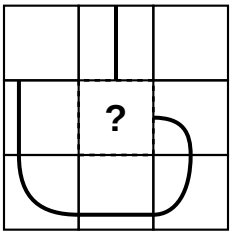
Math Kangaroo Practice - Grades 3-4

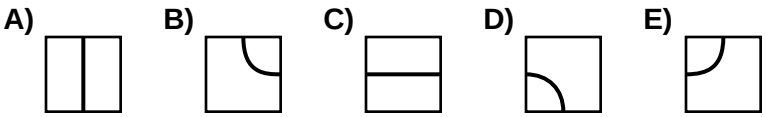
Practice set for 2002 #1, 2002 #4, 2002 #6, 2002 #7, 2003 #9, 2003 #18, 2007 #8
Three practice problems per selected problem.

Set 1 - Missing puzzle tile

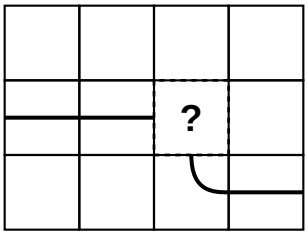
Based on 2002, Problem 1

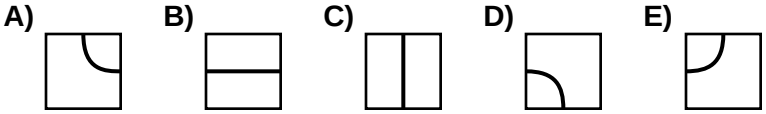
Practice 1.1 A train track runs across the grid below. Which tile belongs in the empty square marked with a question mark (?) to complete the continuous track?



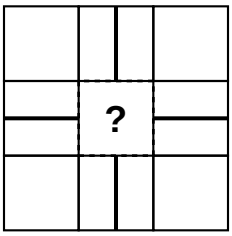


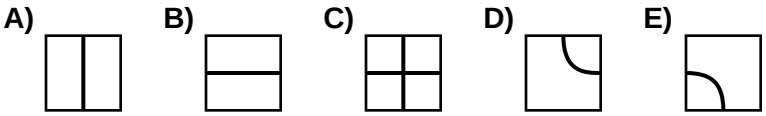
Practice 1.2 A pipe connects across the grid. Which tile is missing at the question mark (?) to connect the left pipe to the bottom pipe?





Practice 1.3 Two roads cross each other on the map below: one road runs vertically and the other runs horizontally. Which tile belongs in the empty square marked with a question mark (?) so both roads can continue across the square?

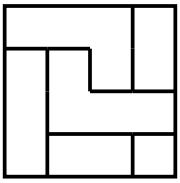


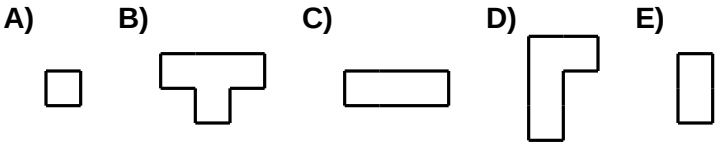


Set 2 - Cutting a square into pieces

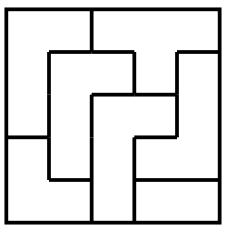
Based on 2002, Problem 4

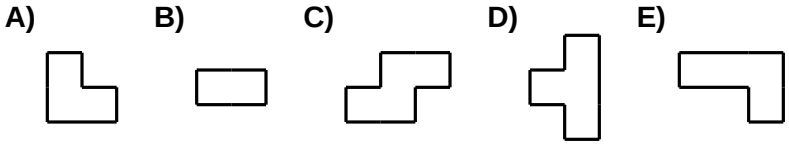
Practice 2.1 The square was cut along the lines. Which of the shapes below was **not** obtained this way?



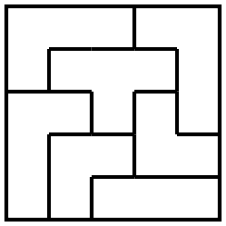


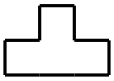
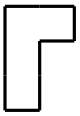
Practice 2.2 This square was cut along the lines. Which shape was **not** obtained?





Practice 2.3 This square was cut along the lines. Which shape was **not** obtained?



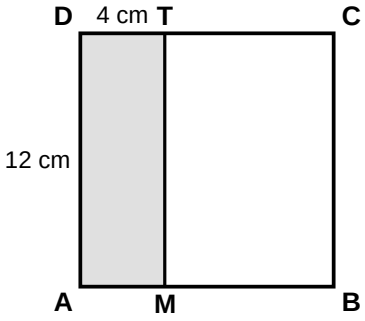
A)  **B)** **C)** **D)**  **E)**

Set 3 - Square and rectangle perimeters

Based on 2002, Problem 6

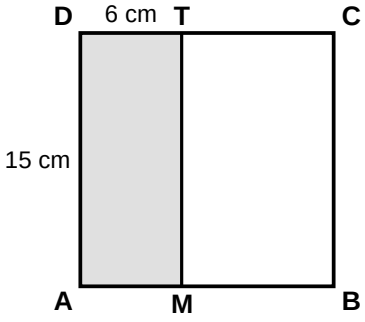
Practice 3.1 The side of square *ABCD* is 12 cm. The side *AM* of rectangle *AMTD* is 4 cm. How much greater is the perimeter of square *ABCD* than the perimeter of rectangle *AMTD*?

A) 10 cm **B)** 16 cm **C)** 18 cm **D)** 20 cm **E)** 24 cm



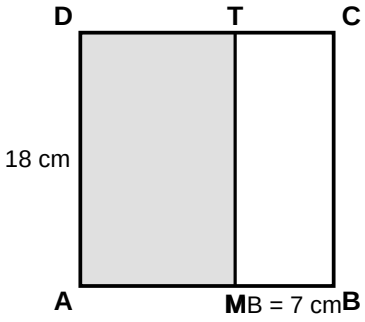
Practice 3.2 The side of square *ABCD* is 15 cm. The side *AM* of rectangle *AMTD* is 6 cm. How much greater is the perimeter of square *ABCD* than the perimeter of rectangle *AMTD*?

A) 14 cm **B)** 16 cm **C)** 18 cm **D)** 22 cm **E)** 24 cm



Practice 3.3 The side of square *ABCD* is 18 cm. A line segment *MT* divides the square into two rectangles. The length of segment *MB* is 7 cm. How much greater is the perimeter of square *ABCD* than the perimeter of rectangle *AMTD*?

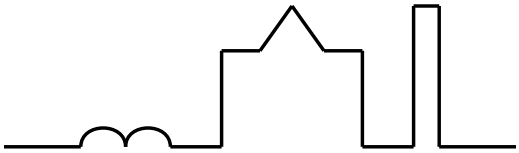
A) 12 cm **B)** 14 cm **C)** 16 cm **D)** 18 cm **E)** 22 cm



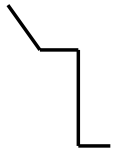
Set 4 - Pieces of a skyline

Based on 2002, Problem 7

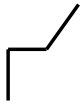
Practice 4.1 Here is the skyline of a town. Which piece **cannot** belong to it?

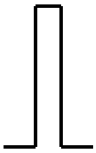


A)

B) 

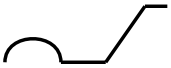
C)

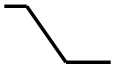
D) 

E) 

Practice 4.2 Here is the skyline of a castle. Which piece **cannot** belong to it?

A)

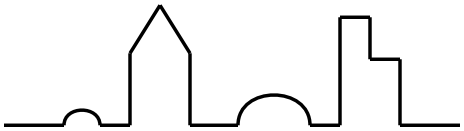
B) 

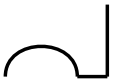
C) 

D)

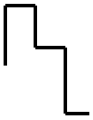
E)

Practice 4.3 Here is a skyline with a small arch, a tower and a dome. Which piece **cannot** belong to it?



A) 

B)

C) 

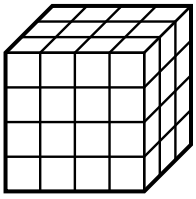
D)

E)

Set 5 - Red outside, blue inside

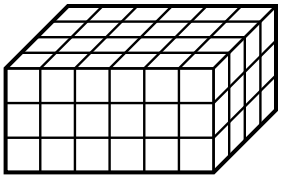
Based on 2003, Problem 9

Practice 5.1 Maya built this brick (4 cubes wide, 4 cubes tall, 3 cubes deep). The outside is completely red and all the cubes inside are blue. How many blue cubes did she use?



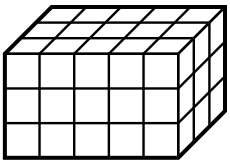
A) 2 **B)** 4 **C)** 6 **D)** 8 **E)** 12

Practice 5.2 Leo built this brick (6 cubes wide, 3 cubes tall, 4 cubes deep). The outside is completely red and all the cubes inside are blue. How many blue cubes did he use?



A) 4 **B)** 6 **C)** 8 **D)** 12 **E)** 16

Practice 5.3 Anna built this brick (5 cubes wide, 3 cubes tall, 3 cubes deep). The outside is completely red and the cubes inside are blue. How many **red** cubes did she use?

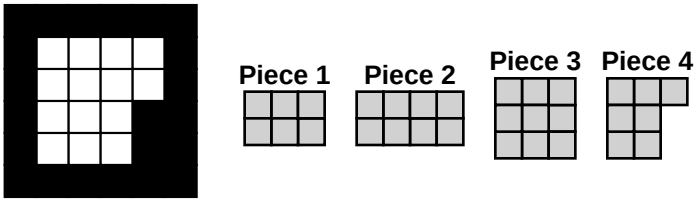


A) 30 **B)** 36 **C)** 39 **D)** 42 **E)** 45

Set 6 - Covering an empty shape with two pieces

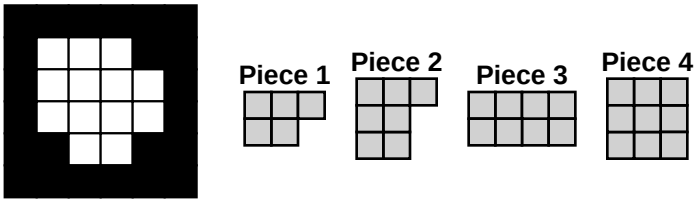
Based on 2003, Problem 18

Practice 6.1 The picture shows a 6 × 6 square with some cells black and the rest white. We want to cover the white area completely without overlaps using two of the numbered pieces (1, 2, 3, 4). Which two pieces should we choose?



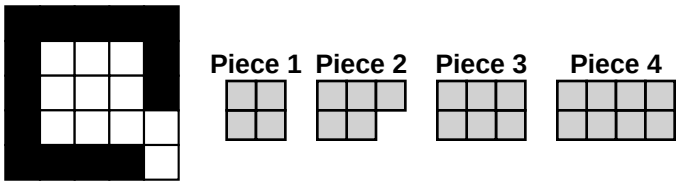
A) 1 and 2 **B)** 1 and 3 **C)** 2 and 4 **D)** 3 and 4 **E)** 2 and 3

Practice 6.2 The picture shows a 6 × 6 square with some cells black and the rest white. We want to cover the white area completely without overlaps using two of the numbered pieces (1, 2, 3, 4). Which two pieces should we choose?



A) 1 and 2 **B)** 2 and 3 **C)** 1 and 3 **D)** 2 and 4 **E)** 3 and 4

Practice 6.3 The picture shows a 5 × 5 square with some cells black and the rest white. We want to cover the white area completely without overlaps using two of the numbered pieces (1, 2, 3, 4). Which two pieces should we choose?

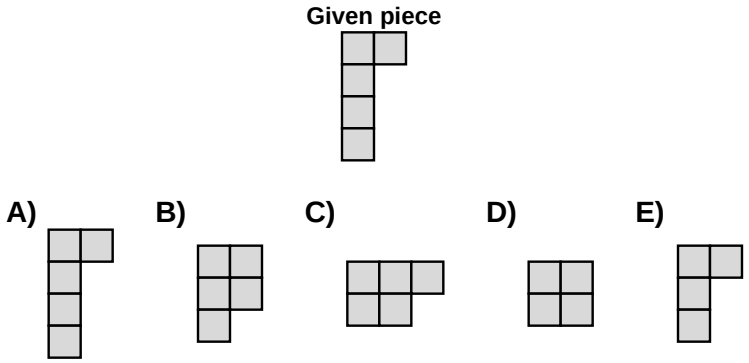


A) 1 and 2 **B)** 1 and 4 **C)** 2 and 4 **D)** 2 and 3 **E)** 3 and 4

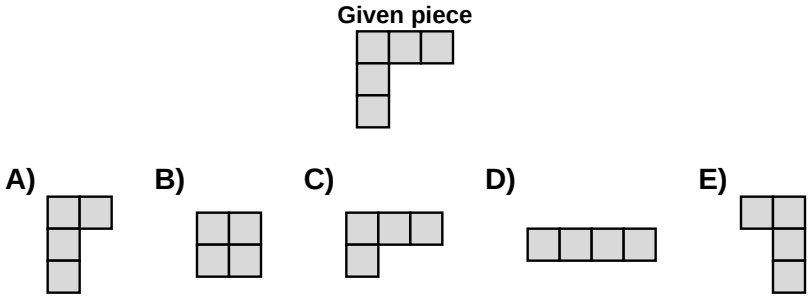
Set 7 - Joining pieces to form a rectangle

Based on 2007, Problem 8

Practice 7.1 Which of the shapes (A through E) can be joined together with the given shape to form a solid rectangle without gaps or overlaps?

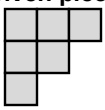


Practice 7.2 Which of the shapes (A through E) can be joined together with the given shape to form a solid square (or rectangle) without gaps or overlaps?

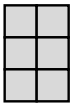


Practice 7.3 Which of the shapes (A through E) can be joined together with the given shape to form a solid rectangle without gaps or overlaps?

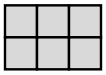
Given piece



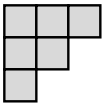
A)



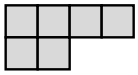
B)



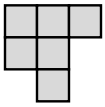
C)



D)



E)



Answer Key & Explanations

For generated/2026-10-19-12-46-33.pdf (Practice set for 2002 #1, 2002 #4, 2002 #6, 2002 #7, 2003 #9, 2003 #18, 2007 #8)

Solutions & kid-friendly explanations

1.1: B The track enters the empty square from the top at the center and continues out to the right. Tile B contains the curved track connecting the top edge to the right edge.

1.2: D The incoming pipe comes from the left at the middle, and exits through the bottom edge at the middle. Tile D correctly turns the pipe from left to bottom.

1.3: C Both the vertical road (top to bottom) and the horizontal road (left to right) pass through the missing square. Tile C shows the intersection of both roads.

2.1: B Pieces: two L-shapes of 4, a straight 3, two dominoes, one small square. No T-shape.

2.2: C Pieces: four L-shapes of 4, one T, one L of 3, one domino. There is no zig-zag (S) piece.

2.3: E Pieces: three L-shapes of 4, one T, two L-shapes of 3, one straight 3. No 2x2 square.

3.1: B Perimeter of square ABCD = 4 × 12 = 48 cm. Perimeter of rectangle AMTD = 2 × (12 + 4) = 32 cm. Difference = 48 - 32 = 16 cm.

3.2: C Perimeter of square ABCD = 4 × 15 = 60 cm. Perimeter of rectangle AMTD = 2 × (15 + 6) = 42 cm. Difference = 60 - 42 = 18 cm.

3.3: B Since AB = 18 cm and MB = 7 cm, AM = 18 - 7 = 11 cm. Perimeter of square = 4 × 18 = 72 cm. Perimeter of rectangle AMTD = 2 × (18 + 11) = 58 cm. Difference = 72 - 58 = 14 cm (which is simply 2 × MB = 2 × 7 = 14 cm).

4.1: A In the skyline there is a flat part between the wall and the pointed roof. Piece A has the roof sitting straight on the wall.

4.2: E The castle wall has TWO notches at the top. Piece E shows a whole tower top with only one notch.

4.3: D After each arch the line goes flat and then UP. Piece D goes down after the arch.

5.1: B Inside is (4-2) x (4-2) x (3-2) = 2 x 2 x 1 = 4.

5.2: C Inside is (6-2) x (3-2) x (4-2) = 4 x 1 x 2 = 8.

5.3: D Total = 5 x 3 x 3 = 45. Inside (blue) = 3 x 1 x 1 = 3. Red = 45 - 3 = 42.

6.1: A Counting the white cells gives 14 empty cells. The pieces have 6, 8, 9, and 7 squares. Only Piece 1 and Piece 2 have 6 + 8 = 14 squares.

6.2: C Counting the white cells gives 13 empty cells. The pieces have 5, 7, 8, and 9 squares. Only Piece 1 and Piece 3 have 5 + 8 = 13 squares.

6.3: D Counting the white cells gives 11 empty cells. The pieces have 4, 5, 6, and 8 squares. Only Piece 2 and Piece 3 have 5 + 6 = 11 squares.

7.1: A The given shape has column heights 4 and 1 (5 squares). Shape A is identical; turned upside-down it has column heights 1 and 4. Joining them gives columns of height 4+1=5 each, creating a 5 × 2 rectangle.

7.2: B The given shape has 5 squares forming a corner inside a 3 × 3 boundary. A 2 × 2 square (shape B, 4 squares) fits into the empty corner to complete a solid 3 × 3 square.

7.3: C The given staircase shape has column heights 3, 2, and 1 (6 squares). Shape C is the same staircase; turned upside-down it has column heights 1, 2, and 3. Joined together, every column has height 4, forming a solid 4 × 3 rectangle of 12 squares.